Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	6
Topic	Construction
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees
Theme	—
Curriculum objectives	• construct parallel lines • construct perpendicular lines • bisect a given line segment • construct angle 90° • construct angle 60°
Previous knowledge	• Students have been taught how to draw lines.
Lesson sequence	• construct parallel lines — the teacher draws the construction of parallel lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student construct parallel lines, following the teacher's board construction and replicating it in their notes • construct perpendicular lines — the teacher draws perpendicular lines on the board step by step, using a set square and ruler and explaining aloud as she draws; the student follow the teacher's board construction and replicate it in their notes • bisect a given line segment — the teacher draws the bisection of a line segment on the board step by step with a pair of compasses, explaining aloud as she draws; the student construct line segment, following the teacher's board construction and replicating it in their notes • construct angle 90° — the teacher constructs angle 90 degrees on the board step by step by compass arcs from a point on the line, explaining aloud as she constructs; the student construct angle 90 degrees, following the teacher's board construction and replicating it in their notes • construct angle 60° — the teacher constructs angle 60 degrees on the board step by step by compass arcs of equal radius from the endpoint, explaining aloud as she constructs; the student construct angle 60 degrees, following the teacher's board construction and replicating it in their notes
Instructional materials	• Mathematical set.
Evaluation	—

Lesson objectives

1.	construct parallel lines
2.	construct perpendicular lines
3.	bisect a given line segment
4.	construct angle 90°
5.	construct angle 60°

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees

Class: JSS1 **Subject:** Mathematics **Week:** 6

Section 1: Constructing a Parallel Line Through a Point

Two lines are parallel when they run side by side and never meet, no matter how far they are extended. A railway track is a good picture: the two rails stay the same distance apart all the way along.

In this section you will construct a line that passes through a given point P and is parallel to a given line AB. You will need a set square and a ruler — the same tools you use for drawing lines. The set square does the work of keeping the new line parallel; the ruler simply holds the set square steady while you slide it.

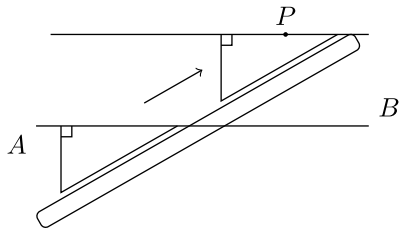
Example: Construct a line through the point P so that it is parallel to AB.

Step (a) — place the set square on the line.
Place a set square so that one of its edges lies exactly along AB (fig-01). Take care to line the edge up with the line the whole way along it. The point P is marked above the line, where the new line must pass.

Step (b) — hold the set square with the ruler.
Place the ruler along one of the set square's other edges (fig-02). If you are right-handed, use the edge on your left; if left-handed, the other one. It does not matter which edge you choose, and the set square can be used either way round, so long as one edge lies on AB and the ruler rests against another edge of it (fig-03). The ruler's slope across the page does not affect the construction — the ruler is only there to hold the set square still.

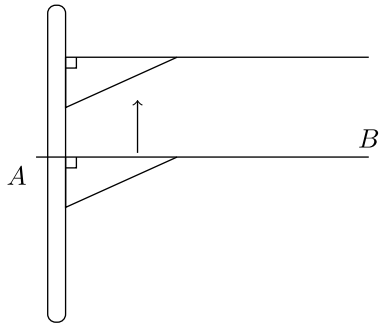
Step (c) — slide the set square to P and draw.
Hold the ruler firmly in place with one hand. With the other hand, slide the set square along the ruler towards P (fig-04). Slide it until the edge that was lying on AB reaches P (fig-05). Because you slid the set square without turning it, this edge stays parallel to AB the whole time. Draw a line along that edge of the set square, through P. That line is parallel to AB.

Visual



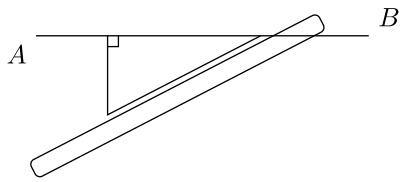
The same slide as fig-04, made against the leaning ruler of fig-03. The set square is drawn twice: once with its edge on AB, and once slid along the ruler until that edge passes through P. The two positions are congruent and point the same way — sliding along a held ruler moves the set square without turning it — so the edge through P stays parallel to AB. The arrow gives the direction of the slide, along the ruler's edge, and the right angle is marked in both positions.

Visual



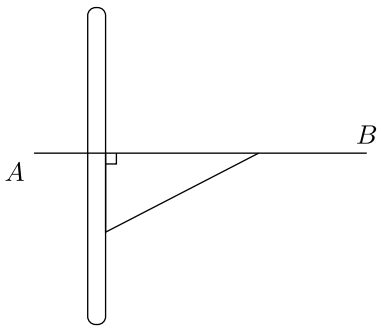
The vertical ruler held fixed, with the set square drawn twice — once with its edge on AB and once higher up, its edge on a second line parallel to AB — and an arrow between the two positions pointing up along the ruler.

Visual



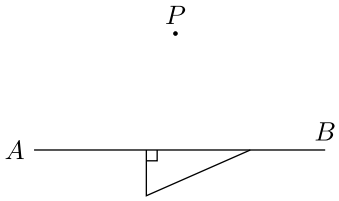
The same set square on AB as in fig-02, with the ruler moved to rest against its sloping edge instead of its upright one. The set square has not moved; only the ruler has, so it now leans across AB, running from lower left to upper right. The right angle is still where the edge on AB meets the short upright edge, and it is marked there. The ruler's inclination does not affect the construction. The set square is drawn at one particular size, and with its right angle at one particular end; neither is part of the method. Any set square works, either way round, so long as one edge lies along AB and the ruler rests against another.

Visual



The same set square on AB, with the ruler now drawn as a long strip lying along the set square's downward edge. The right angle marked at the corner is the set square's own, between the edge lying along AB and the edge running down from it.

Visual



Line AB, with a point P marked above it. A set square is drawn below AB with one edge along the line: the right angle is at the left end of that edge, one leg runs down from it, and the hypotenuse closes back to a point on AB to the right.

Practice

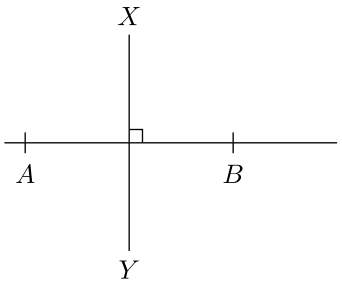
1. Describe, in your own words, the steps you would take to construct a line through a point P that is parallel to a line AB.
2. While you slide the set square along the ruler towards P, why must you hold the ruler firmly and slide the set square without turning it?

Section 2: Constructing Perpendicular Lines

Two lines are said to be **perpendicular** to each other when they meet at a right angle — that is, at an angle of $90\,^\circ$. In fig-06, the line XY crosses the line AB, and the small square mark in the upper right corner of the crossing tells you the two lines meet at a right angle. Wherever you see that little square, the lines are perpendicular.

You can draw a perpendicular line yourself using just a set square and a ruler. A set square is a flat triangular piece of plastic with one corner that is exactly a right angle, and the edge that faces that corner is called the hypotenuse. Here is how to use it.

Visual



Line AB with a second line XY crossing it, X above and Y below, and a right-angle mark in the upper right quadrant at the crossing.

Construct a line perpendicular to line AB through a point Q on it.

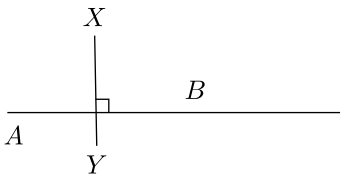
Step 1. Draw a line AB, as in fig-07, and mark a point Q on the line.

Step 2. Place the edge of the set square on the line AB, as shown in fig-08. Let the edge of the ruler meet the hypotenuse of the set square. The set square stands on the line with one upright edge rising from it and a marked right angle there.

Step 3. Pressing firmly on the ruler so it does not slip, slide the set square down along the hypotenuse side until the third edge touches the point Q. Then draw a line XY along that edge.

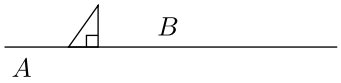
The line XY you have drawn crosses AB at point Q, and the right-angle mark at the crossing (fig-09) shows the two lines meet at $90\,^{\circ}$. So XY is perpendicular to AB.

Visual



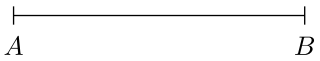
Line AB with XY now drawn across it, X above and Y below, and a right-angle mark in the upper right quadrant.

Visual



Line AB with the set square standing on it as a narrow right-angled triangle: a short base along AB, the upright edge rising from the right end of that base with the right angle marked there, and the hypotenuse closing back down to the left.

Visual



A plain line with a stroke at each end, labelled A on the left and B on the right.

Practice

1. Two lines are perpendicular when they meet at what angle?
2. Draw a line AB and mark a point Q on it. Using your set square and ruler, construct a line through Q that is perpendicular to AB. What small mark should you add at the crossing to show the lines are perpendicular?

Section 3: Bisecting a Line Segment

To bisect a line segment means to divide it into two equal halves. When you bisect the segment AB, you find its midpoint M, the point exactly halfway along it, so that AM = MB. You do this with a pair of compasses and a ruler.

Bisect the line segment AB.

Step 1. Open your pair of compasses so that the radius is about ³/₄ of the length of AB.</p>

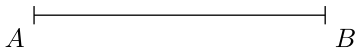
Step 2. Place the sharp point of the compasses on A. Draw two arcs — one above and the other below the middle of AB.</p>

Step 3. Keeping the same radius, place the sharp point on B. Draw two arcs so that they cut the first arcs at P (above) and Q (below).</p>

Step 4. Join P to Q with a ruler.</p>

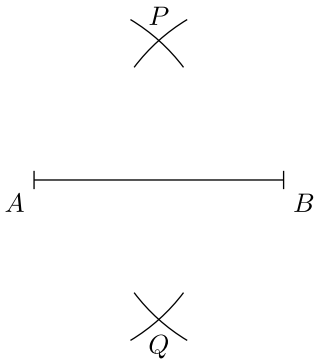
The line PQ crosses AB at M. M is the midpoint of AB, so AM = MB. Notice also that PQ meets AB at 90° — the two lines cross at a right angle.</p>

Visual



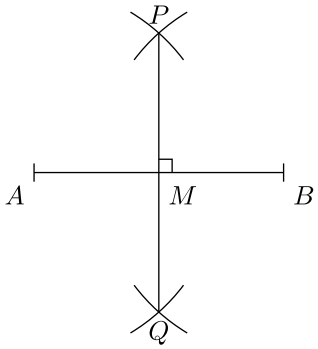
The line segment AB, a stroke at each end, A on the left and B on the right.

Visual



The segment AB with arc pairs crossing above and below it, the upper crossing labelled P and the lower one Q. Each crossing is drawn as a small X where an arc from A meets an arc from B.

Visual



The same arcs with P joined to Q, the joining line crossing AB at its middle at M and carrying a right-angle mark in the upper right quadrant.

Practice

1. Draw a line segment CD and bisect it with your compasses and ruler. What can you say about the two parts of CD after you have bisected it?
2. When you bisect the segment AB, the line PQ crosses AB at M. Write down the relationship between AM and MB, and the angle at which PQ meets AB.

Section 4: Constructing a 90° Angle

To construct a 90° angle you only need a ruler and a pair of compasses. You build the perpendicular on an existing line, and where a perpendicular meets the line it makes a right angle, which measures 90°.

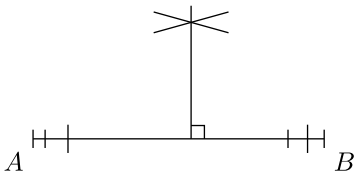
Begin by drawing a straight line AB and marking a point C on it. This is the point where the 90° angle is to be made, and it is where the perpendicular will stand.

1. With the compasses resting at C, draw an arc that crosses the line at two points on either side of C. Call those points D and E. (See fig-13: the two short strokes on the line towards A and the two towards B mark where these arcs touch the line.)

2. Now open the compasses WIDER than the distance DC. This matters: the width you used in step 1 will not work, because two arcs drawn at that width from D and E would meet only at C itself.
3. With the compasses at this new, wider setting, place the point at D and draw an arc above the line.
4. Keeping the compasses at that same wider setting, move the point to E and draw another arc, cutting the first one at a point F.
5. Join C to F with a ruler.

The line CF is perpendicular to AB, so the angle between CF and the line — written ∠FCB — is 90°.

Visual



A line with a stroke at each end, *A* on the left and *B* on the right, and four short strokes standing on the line, a pair towards *A* and a pair towards *B*. A perpendicular rises from a point near the middle, with a right-angle mark at its foot, and two short strokes cross each other near its top.

Constructing ∠FCB = 90°

Draw the line AB and mark point C on it. With the compasses at C, swing one arc to cut the line at D and E on either side of C. Now open the compasses wider than DC. From D, draw an arc above the line. Without changing the compass width, move to E and draw another arc that cuts the first at F. Join C to F with a straight line.

The result looks like fig-13: a perpendicular rises from a point on the line, marked with a small right-angle square at its foot, and two short arcs cross near its top where the two wider arcs met. Because CF is perpendicular to AB, the angle ∠FCB is exactly 90°.

Practice

1. When you construct a 90° angle on a line AB at point C, you first draw an arc from C that cuts the line at D and E. Then you must open the compasses wider than DC before drawing the arcs from D and E. Why is the wider opening necessary?
2. A line AB is drawn with point C marked on it. To construct a right angle at C, name, in the correct order, the four compass steps you take, and say what you draw last with a ruler.

Section 5: Constructing an Angle of 60°

Here you will learn to construct an angle of exactly 60° using only a ruler and a compass. The secret is to swing two arcs of the same radius, because an equilateral triangle has all three angles equal to 60°.

You already know how to draw a straight line. Now you need your compass set to one convenient radius and kept at that same radius for both arcs.

Construct angle 60° at point B on line BC.

Step 1 — Draw the base line. Draw a line segment BC, with B at the left end and C further along to the right. Set your compass to a convenient radius.

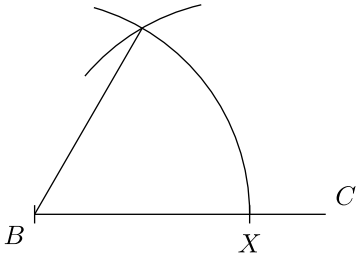
Step 2 — First arc. Put the compass point on B (the centre) and draw an arc that cuts BC at a point X. Keep the compass at this same radius — do not change it.

Step 3 — Second arc. Put the compass point on X (your new centre) and, with the *same radius*, draw another arc that cuts the first arc. Call the point where the two arcs meet A.

Step 4 — Join. Use your ruler to draw the straight line from B to A.

The angle between BA and BC at B is exactly 60°. This works because in triangle ABX, all three sides BX, XA and AB are equal — they are all the same compass radius — so it is an equilateral triangle, and every angle of an equilateral triangle is 60°.

Visual



The line BC with B at the left, a stroke marking X part way along, a long arc swung from B through X and up to the left, a shorter arc swung from X cutting it, and a straight line from B to that crossing.

Practice

1. Construct an angle of 60° on a fresh line, working carefully through the four steps. When you have joined the last line, use your protractor to check that the angle at the endpoint really is 60°.

Check your work

Attempt every question before you read this page.

Constructing a Parallel Line Through a Point

1. Place a set square with one edge exactly along AB. Hold it steady by placing the ruler along another edge of the set square. Holding the ruler firmly, slide the set square along the ruler until the edge that was on AB reaches P. Draw a line along that edge through P — it is parallel to AB.
2. Holding the ruler firmly keeps the slide straight. Sliding without turning the set square keeps its edge at the same angle as AB, so that edge stays parallel to AB — and the line you draw through P is therefore parallel to AB.

Constructing Perpendicular Lines

1. 90 ° (a right angle)
2. You should add a small right-angle square in one quadrant of the crossing, showing the lines meet at 90 °.

   Place the edge of the set square on line AB, let the ruler meet the hypotenuse, slide the set square along until its third edge touches point Q, and draw line XY. Mark the little square at the crossing.

Bisecting a Line Segment

1. The bisecting line cuts CD at its midpoint, so the two parts are equal in length — you can write CM = MD, where M is the midpoint.

   Open the compasses to about three-quarters of the length of CD. Draw two arcs from C (one above, one below), then keep the same radius and draw two arcs from D that cut them at P and Q. Join P to Q; where it crosses CD is the midpoint M, so CM = MD.
2. AM = MB, and PQ meets AB at 90° (a right angle).

Constructing a 90° Angle

1. If the compasses were left at the width DC, the two arcs drawn from D and E would meet only at C itself, so they would not give you a point F above the line to join to C. Opening wider makes the two arcs cut each other at a point F above the line.
2. a. With the compasses at C, draw an arc that cuts the line at D and E.
   b. Open the compasses wider than DC, and from D draw an arc above the line.
   c. Keeping the same wider width, from E draw another arc to cut the first at F.
   d. Join C to F with a ruler.

   The order matters. First fix the two cut points D and E on the line with one arc from C. Then, with a wider compass opening (wider than DC), draw an arc from D above the line, keep the width, and draw a second arc from E that crosses the first at F. Finally use the ruler to draw CF, which is perpendicular to AB and makes ∠FCB = 90°.

Constructing an Angle of 60°

1. Your constructed angle should measure 60° when checked with a protractor. If it does not, repeat the construction keeping the compass radius the same for both arcs — changing the radius between the arcs is the most common reason the angle is wrong.

   Draw a base line BC with B on the left. With B as centre, swing an arc to cut BC at X. Keeping the compass at the same radius, swing a second arc with X as centre to cut the first arc at A. Join B to A. Triangle ABX is equilateral because AB = BX = XA, so angle ABC = 60°.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-b0129758	**Date**	
Subtopic	Construction: parallel and perpendicular lines, bisection, and angles 90 and 60 degrees	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____